US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12416550
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Wanders; Bart et al.

Sampling system

Abstract

A sampling system that includes a traverse, a sample aspiration module body, and a mechanical system. The sample aspiration module body is configured to travel along the traverse. The mechanical system is configured to move the traverse from a first position to a second position, wherein both the first position and the second position lie in the same plane.

Inventors: **Wanders; Bart (Trabuco Canyon, CA), Grace; Eric (Deland, FL)**

Applicant: **Beckman Coulter, Inc.** (Brea, CA)

Family ID: **1000008818967**

Assignee: **Beckman Coulter, Inc. (Brea, CA)**

Appl. No.: **18/109604**

Filed: **February 14, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230258540 A1	Aug. 17, 2023

Related U.S. Application Data

us-provisional-application US 63309887 20220214

Publication Classification

Int. Cl.: **G01N35/10** (20060101); **G01N1/14** (20060101)

U.S. Cl.:

CPC **G01N1/14** (20130101); G01N2001/1418 (20130101)

Field of Classification Search

CPC: G01N (1/14); G01N (2001/1418); G01N (35/1083); G01N (2035/0406); G01N (35/04); G01N (2001/1427); G01N (35/109); G01N (2035/0401)

USPC: 436/180; 422/63; 422/509; 73/53.01; 73/61.55; 73/61.59; 73/64.56; 73/863; 73/863.01; 73/864.24; 73/864.25

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3687632	12/1971	Natelson	422/510	G01N 35/1083

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1275966	12/2002	EP	B01L 9/543
WO-2013174961	12/2012	WO	B01F 29/31
WO-2018038019	12/2017	WO	B01J 4/02
WO-2018173464	12/2017	WO	B01L 3/523

Primary Examiner: Raevis; Robert R

Attorney, Agent or Firm: Frost Brown Todd LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a non-provisional of, and claims the benefit of, provisional patent application 63/309,887, titled “Sampling System” and filed in the USPTO on Feb. 14, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE
(1) Embodiments of the present disclosure relate generally to a system that provides specimen tube analysis. Specific embodiments provide transportation of specimen tubes for presentation, mixing, sampling, and/or analysis in various preparation chambers along a single plane.
BACKGROUND
(2) Some systems provide specimen tube analysis and movement through an analysis system that generally moves specimen tubes along multiple planes or axes in order to accommodate necessary functions. Although these systems are functional, the need for motion and multiple axes adds to the complexity and cost of the mechanism required. The use of multiple axes can also reduce reliability, leading to potentially increased downtime for repairs.
BRIEF SUMMARY
(3) Accordingly, this disclosure relates to a system that provides specimen tube analysis using only a single plane of motion. It reduces the overall complexity and cost of system, while also increasing reliability. Embodiments of the present disclosure provide a sampling system with components that move along a single plane. In some embodiments, the components of a sampling system are

contained in a vertical orientation in order to have the components along a single plane.
(4) In some embodiments, a sampling system includes a traverse which supports a sample aspiration module body and an aspiration probe, a cassette holding one or more specimen tubes, and one or more preparation chambers for sample analysis whereby the module body and aspiration probe are configured to deposit the contents of the specimen tubes into the preparation chambers. In some embodiments, these elements are contained within a single plane or located along a single axis. In some examples, these elements are horizontally and/or vertically offset from each other but still contained within a single plane or along a single axis.
(5) In one embodiment, a traverse used as part of a sampling system has two degrees of movement (e.g., horizontal and vertical) but these are linked movements and the traverse is limited to movement along a single plane or single axis. In some embodiments, the traverse is movable and in some embodiments the traverse is fixed. In some examples, the traverse is connected to one or more cams or linked to an angled track mechanism.
(6) In some embodiments, a sampling system utilizes an angled traverse and a sample aspiration module mounted to the angled traverse, where various components of the sampling system and vertically offset from each other. In one example, the vertical offset comprises a staggered configuration.
(7) In some embodiments, a sampling system utilizes a traverse, a sample aspiration module body configured to travel along the traverse, and one or more specimen tubes and one or more preparation chambers where the one or more specimen tubes and one or more preparation chambers are vertically offset along a same or similar plane.
(8) In some embodiments, methods of operating a sampling system (such as the sampling system embodiments described above) are contemplated. In some embodiments, a method comprises moving a traverse along a plane and moving a sample aspiration module body linked to the traverse. In some embodiments, a method comprises moving a sample aspiration module body along an angled traverse to access different regions of a sampling system (e.g., a specimen tube, and preparation chambers). In some embodiments, a method comprises utilizing a sample aspiration module body to take a sample from a specimen tube, moving the sample aspiration module body along a traverse to deposit the sample into a preparation chamber, wherein the specimen tube and the preparation chamber are vertically offset along a same or similar plane.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A shows a schematic front plan view of an exemplary sampling system that moves in a single plane.
(2) FIG. **1**B shows a schematic front plan view of the sampling system of FIG. **1**A, but with a sample aspiration module body positioned over a first specimen tube and a probe extending into the tube to collect a sample.
(3) FIG. **1**C shows a schematic front plan view of the sampling system of FIG. **1**B, but with the probe retracted.
(4) FIG. **1**D shows a schematic front plan view of the sampling system of FIG. **1**C, but with a mechanical system moving the traverse to an upper (second) position, such that the sample aspiration module body is positioned over a mix chamber.
(5) FIG. **1**E shows a schematic front plan view of the sampling system of FIG. **1**D, but with the sample aspiration module body positioned over a mix chamber and with the probe extending into the mix chamber to deliver the sample.
(6) FIG. **1**F shows a schematic front plan view of the sampling system of FIG. **1**E, but with the probe retracted from the first mix chamber.

(7) FIG. **1**G shows a schematic front plan view of the sampling system of FIG. **1**F, but with the sample aspiration module body positioned over a second mix chamber.
(8) FIG. **1**H shows a schematic front plan view of the sampling system of FIG. **1**G, but with the probe extending into the second mix chamber to deliver the sample.
(9) FIG. **1**I shows a schematic front plan view of the sampling system of FIG. **1**H, but with the probe retracted from the second mix chamber.
(10) FIG. **1**J shows a schematic front plan view of the sampling system of FIG. **1**I, but with the mechanical system moving the traverse to a first position, such that the sample aspiration module body is positioned over a second specimen tube.
(11) FIG. **2** shows a schematic side plan view of the sampling system of FIG. **1**B.
(12) FIG. **3** shows a schematic top view of the sampling system of FIG. **1**B.
(13) FIG. **4**A shows a schematic front plan view of the sampling system of FIG. **1**A, but with the traverse locating the sample aspiration module body in the first position.
(14) FIG. **4**B shows a schematic front plan view of the sampling system of FIG. **4**A, but with the mechanical system having moved the traverse to a second position.
(15) FIG. **4**C shows a schematic front plan view of the sampling system of FIG. **4**B, but with the sample aspiration module body positioned over a manual sampling position.
(16) FIG. **4**D shows a schematic front plan view of the sampling system of FIG. **4**C, but with the probe extended into the manual sampling specimen tube.
(17) FIG. **4**E shows a schematic front plan view of the sampling system of FIG. **4**D, but with the probe retracted from the manual sampling specimen tube.
(18) FIG. **4**F shows a schematic front plan view of the sampling system of FIG. **4**E, but with the sample aspiration module body positioned over a mix chamber.
(19) FIG. **4**G shows a schematic front plan view of the sampling system of FIG. **4**F, but with the probe extended into the mix chamber.
(20) FIG. **5**A shows a schematic front plan view of the sampling system of FIG. **1**A, but with the traverse locating the sample aspiration module body in the first position.
(21) FIG. **5**B shows a schematic front plan view of the sampling system of FIG. **5**A, but with the mechanical system having moved the traverse to a second position.
(22) FIG. **5**C shows a schematic front plan view of the sampling system of FIG. **5**B, but with the sample aspiration module body positioned over an open vial sampling position.
(23) FIG. **5**D shows a schematic front plan view of the sampling system of FIG. **5**C, but with the probe extended into the open vial.
(24) FIG. **5**E shows a schematic front plan view of the sampling system of FIG. **5**D, but with the probe retracted.
(25) FIG. **5**F shows a schematic front plan view of the sampling system of FIG. **5**E, but with the sample aspiration module body positioned over a mix chamber.
(26) FIG. **5**G shows a schematic front plan view of the sampling system of FIG. **5**F, but with the probe extended into the mix chamber.
(27) FIG. **6**A shows a schematic front plan view of an exemplary sampling system that includes an angled traverse, with the sample aspiration module body positioned over a first specimen tube.
(28) FIG. **6**B shows a schematic front plan view of the sampling system of FIG. **6**A, but with the probe of the sample aspiration module body extended into the first specimen tube to collect a sample.
(29) FIG. **6**C shows a schematic front plan view of the sampling system of FIG. **6**B, but with the sample aspiration module body moved up the angle traverse and positioned over a first mix chamber.
(30) FIG. **6**D shows a schematic front plan view of the sampling system of FIG. **6**C, but with the probe of the sample aspiration module body extended into first mix chamber to deliver a sample.

DETAILED DESCRIPTION

(31) Existing sampling systems utilize complicated multi-planar movements where different components of the system must be able to navigate in various directions along different axes and different planes in order to obtain a sample. This results in complexity, increased risk of mechanical failure, and also can occupy significant space creating a resource burden. Such existing systems are described, for instance, in U.S. Pat. Nos. 7,331,474, 7,028,831, 7,850,914, the disclosure of each of the above-cited U.S. Patents is incorporated by reference herein in its entirety.
(32) For the purposes of the Figures, FIGS. **1**A-**1**J, **4**A-**6**D generally show a frontal plan view of the various systems described herein. As such, there is an x-axial direction extending horizontally (e.g., left to right), a y-axial direction extending vertically (e.g., up and down), and a z-axial direction extending into the page (e.g., depth wise) which generally is not visible since the drawings generally offer a frontal, 2-dimensional view.
(33) Embodiments of the present disclosure provide exemplary sampling systems **10**, **210** that include an aspirator module body **12** that moves along a traverse **14**, **50**. The sampling systems **10**, **210** moves the sample aspiration module body **12** in only a single plane.
(34) The sampling system **10** includes a sample aspiration module body **12** that is positioned on a traverse **14**, with the traverse **14** extending at least a portion of the length L within the system. The sample aspiration module body **12** may have an opening therethrough that receives the shaft of the traverse **14**. Alternatively, the sample aspiration module **12** may be slidably mounted to the traverse **14**. Other securement/movement/attachment possibilities between the sample aspiration module body **12** and the traverse **14** are envisioned within the scope of this disclosure. As will be described further below, the traverse **14** is raised or lowered (in the same plane) via a mechanical system **16**. Raising and lowering of the traverse **14** causes raising and lowering of the sample aspiration module body **12**.
(35) A plurality of specimen tubes **18**, each containing a sample, are positioned on a cassette assembly **20**. The cassette assembly **20** includes individual cassettes **20***a*, **20***b*. The cassettes **20***a*, **20***b* may be moved through the sampling system **10** via a sample transport module **30**. For example, the sample transport module **30** may include a conveyor belt. One or more of the cassettes **20***a*, **20***b* may be associated with a mixer **28** (see FIGS. **2**-**3**), which functions to tilt the cassettes **20***a*, **20***b* (and their respective specimen tube(s) **18**) to maintain the mixed specimen and to prevent any precipitation that may otherwise occur. When the cassette **20***a* is in the desired position with respect to the sample aspiration module body **12**, the sample aspiration module body **12** is positioned over a specimen tube **18**, as illustrated by FIG. **1**A.
(36) The sample aspiration module body **12** supports an aspirator probe **22** which is lowered relative to the sample aspiration module body **12** into a specimen tube **18***a* of a specimen tube arrangement **18** to aspirate a sample from the specimen tube **18**. This raising and lowering movement of the aspirator probe **22** is shown as an up-and-down (i.e., vertically oriented) motion (into and out of the sample aspiration module body **12**). The sample aspiration module body moves along traverse **14** (i.e., longitudinally along traverse **14**) to engage the various specimen tubes **18**, and then the aspirator probe **22** lowers (i.e., vertically moves down from the sample aspiration module body **12**) to enter and exit each specimen tube. FIG. **1**A shows the sample aspiration module body **12** in position over one of the specimen tubes **18***a*-**18***e* and the cassette **20***a*. FIG. **1**B shows the aspirator probe **22** extended from the sample aspiration module body **12** and aspirating a sample from an individual tube **18***a* of specimen tube arrangement **18**. Note, aspirator probe **22** has a retracted configuration (e.g., FIG. **1**A) where probe **22** is contained within sample aspiration module body **12** and an extended configuration (e.g., FIG. **1**B) where probe **22** extends vertically from sample aspiration module body **12**. In FIG. **1**C, the aspirator probe **22** has retracted back into the sample aspiration module body **12**. The sample aspiration module body **12** may provide a cleaning function to clean the outer portion of the aspiration probe **22** so that it does not contaminate the next vial/tube/preparation chamber into which the aspiration probe **22** is inserted.
(37) Alternative embodiments can utilize aspiration probe **22** in a non-retractable position such that

it adopts a constant elongated position relative to sample aspiration module body **12**. In one example, sample aspiration module body **12** can move along a y-direction (i.e., vertically) relative to traverse **14** to enter and exit specimen tubes **18** and preparation chambers **24**. In another example, aspiration probe **22** has a fixed elongated configuration and movement of traverse **14** controls the position of aspiration probe **22** relative to specimen tubes **18** and preparation chambers **24**.

(38) Once the sample has been collected, the sample may be transported to preparation assembly **24**. As shown, the preparation assembly **24** lies in the same plane (e.g., along the same z-axis location) as the cassettes **20***a*, **20***b* of cassette assembly **20**. In other words, the preparation assembly **24** is positioned above and/or to the left or right of the cassette assembly **20**, but not forward or back of the cassettes **20***a*, **20***b* (e.g., not displaced along a z-axis). This is different than prior art systems where a preparation assembly is inwardly displaced (meaning, displaced along a z-axis) relative to the cassettes. A mechanical system **16** moves the sample aspiration module body **12** vertically up and horizontally over to the location of the preparation assembly **24**. As illustrated by FIG. **1**D, the traverse **14** may be secured to the mechanical system **16** that includes cams **26***a*, **26***b*. It should be understood, however, that a single cam (or more than two cams) may be used if desired. As shown in FIGS. **1**A-**1**J, the cams **26***a*, **26***b* may function as rollers or wheels that can rotate to one or more optional hard stops **46**. The hard stops **46** (one shown) may be associated with predetermined positions of the traverse **14** relative to one or more of the specimen tubes **18***a*-**18***j* of the tube arrangement **18**, the preparation chambers/baths **24***a*-**24***d* of the preparation assembly **24**, or the sampling holder **40**. As shown, the cams **26***a*, **26***b* may work collectively together to move the traverse **14** between the predetermined positions. The hard stops **46** of the cams **26***a*, **26***b* may be disposed slightly past the lowest point of the traverse **14**, so that the sample aspiration module body **12** can use the hard stops to 'push off against' and generate a sufficient force to pierce the thick rubber stopper of the specimen tubes **18***a*-**18***j*. In some versions, this may eliminate the need for a large strong motor to drive/position the cams **26***a*, **26***b*. As shown, the cams **26***a*, **26***b* are rotatable, such that rotation of the cams **26***a*, **26***b* causes related movement of the traverse **14**.

(39) When the traverse **14** is raised to its second position as shown by FIG. **1**D, the sample aspiration module body **12** is positioned above the preparation assembly **24**. The body **12** may then be moved to a position to deliver the collected sample to the preparation assembly **24**. While the preparation assembly **24** is shown as including four individual chambers/baths **24***a*-**24***d*, it should be understood that more or fewer preparation chambers/baths may be used, including a single preparation chamber or bath. As shown, the preparation assembly **24** may include an imaging white blood cell (WBC) chamber **24***a*, an imaging red blood cell (RBC) chamber **24***b*, a complete blood count (CBC) white blood count (WBC) bath **24***c*, and a CBC/RBC bath **24***d*.

(40) FIG. **1**E illustrates the aspiration probe **22** being extended into the imaging WBC chamber **24***a*. FIG. **1**F illustrates the aspiration probe **22** retracted into the sample aspiration module body **12**. FIG. **1**G illustrates the sample aspiration module body **12** moved laterally along the traverse **14** to chamber **24***b*. FIG. **1**H illustrates the aspiration probe **22** being extended into the imaging RBC chamber **24***b*. FIG. **1**I illustrates the aspiration probe **22** being retracted into the sample aspiration module body **12**.

(41) This similar movement may continue as the sample aspiration module body **12** is moved laterally along the traverse **14** to additional mixing chambers or baths of preparation assembly **24** (so in this example to baths **24***c*, **24***d*). Once the collected sample has been deposited in the chambers/baths **24***a*-**24***d*, the mechanical system **16** may be reversed in rotation so that the one or more cams **26** move the traverse **14** back down to the first position. In other words, lowering of the traverse **14** positions the sample aspiration module body **12** back to its lowered position (i.e., to the position shown in FIGS. **1**A-**1**C). The next specimen tube **18***b* on the cassette **20** may then be sampled. FIG. **1**J illustrates the sample aspiration module body **12** in position over specimen tube **18***b*, as the aspiration probe **22** is prepared to collect a sample therefrom. The same process as

described above, with the mechanical system **16** moving the traverse **14** back up to the level of preparation assembly **24**. Once each of the desired specimen tubes **18***a*-**18***e* on cassette **20***a* have been sampled, the sample transport module **30** may function similar to a conveyor to move the first cassette **20***a* along within the sampling system **10**, such that the cassette **20***b* may be positioned beneath the sample aspiration module body **12**. Thereafter, the sampling system may process the specimen tubes **18***f*-**18***j* contained in the cassette **20***b* in a similar manner to the specimen tubes **18** of the cassette **20***a*.

(42) FIG. **2** illustrates a side plan view of the sampling system **10**, and FIG. **3** illustrates a top plan view of the sampling system **10**. With respect to FIG. **2**, as this is a side plan view, there is an x-axial direction extending into the page, a y-axial direction extending vertically, and a z-axial direction extending left-to-right. With respect to FIG. **3**, as this is a top plan view, there is an x-axial direction extending top-to-bottom, a y-axial direction extending into the page, and a z-axial direction extending left-to-right.

(43) As shown in FIGS. **2**-**3**, the sample aspiration module body **12**, the specimen tubes **18***a*-**18***j* in the cassette assembly **20**, and the preparation chambers/baths **24***a*-**24***d* are aligned in the same plane P. In other words, all these items are located along the same z-axial position, or along the same plane which extends horizontally (i.e., in the x-direction) along the same z-axial position. Having the sample aspiration module body **12**, the specimen tubes **18***a*-**18***j* on the cassette assembly **20**, and the preparation chambers/baths **24***a-d* aligned in the same plane P allows the sample aspiration module body **12** to travel along the traverse **14** in a single longitudinal direction and for the aspiration probe **22** to collect samples from the specimen tubes and deliver the samples to the preparation assembly **24**. As shown in FIGS. **2**-**3**, the sampling system **10** includes at least one analytical module **32** configured to analyze the samples. The mechanical system **16** may be coupled with the sampling system **10** using a support **34**. Similarly, the preparation chambers/baths **24***a*-**24***d* may be coupled with the sampling system **10** using a support **36**. Specimen tube arrangement **18** and the cassette assembly **20** may be supported on a platform **38**.

(44) In use, the cassettes **20***a*, **20***b* travel along a sample transport module **30** and then into a mixer **28**. In some versions, the cassettes **20***a*, **20***b* may travel on a conveyor belt system on which the mixer **28** is mounted. As shown in FIG. **3**, mixer **28** may be laterally offset from (e.g., displaced along a z-direction from/inward relative to) transport module **30**. The mixer **28** tilts the cassettes to mix the specimens contained within the specimen tubes **18**. In some versions, the mixer **28** may tilt the entire cassette away from the line of travel so that a new cassette (e.g., cassette **20***a*, **20***b*) may be added to allow the sampling system **10** to mix and clear at the same time. This also allows the sampling system **10** to add an urgent/stat sample to be tested in between a regular sample run. The mixer **28** may mix the specimen tubes **18***a*-**18***j*.

(45) FIGS. **4**A-**4**G illustrates a frontal plan view of sampling system **10** using a manual sampling holder **40** that includes a manually-placed specimen tube **42**. The mechanical system **16** moves the sample aspiration module body **12** between a first position **200** (illustrated as a lower position of FIG. **4**A) and a second position **202** (illustrated as an upper position of FIG. **4**B). In this example, rather than aspirating a sample from a specimen tube **18***a*-**18***j* on the cassette assembly **20**, the sample aspiration module body **12** bypasses the preparation assembly **24** and moves to aspirate a sample from the manually-placed specimen tube **42** on holder **40**, shown by FIG. **4**D. Once the sample is aspirated as shown by aspiration probe **22** in FIG. **4**D, the aspiration probe **22** is retracted (see FIG. **4**E), and the sample aspiration module body **12** moves to the preparation chambers/baths **24***a*-**24***d* (see FIG. **4**F). Aliquots of the collected sample may then be deposited into the various chambers/baths **24***a*-**24***d* of the preparation assembly **24** as outlined above, with FIG. **4**G illustrating just the initial aspiration probe **22** extension into the preparation chamber **24***a*.

(46) FIGS. **5**A-**5**G illustrate the sampling system **10** using a manual open vial holder **44**. For example, the manual open vial holder **44** may be for pediatric testing, where the specimen volume is less than typical, due to smaller patient size. Smaller specimen tubes may also be used. Using the

manual open vial holder **44** may allow the user to have direct and free access to the aspiration probe **22** instead of aspirating automatically. As described above, the mechanical system **16** functions similarly to move the sample aspiration module body **12** between a first position **200** (illustrated as a lower position of FIG. **5**A) and a second position **202** (illustrated as an upper position of FIG. **5**B). The sample aspiration module body **12** may bypass the chambers/baths **24***a*-**24***d* as well as the manual sampling holder **40** and move to aspirate a sample from a manually-placed specimen tube using holder **44**, shown in FIG. **5**C. Once the sample is aspirated as shown by aspiration probe **22** in FIG. **5**D, the aspiration probe **22** is retracted as shown in FIG. **5**E, and the sample aspiration module body **12** is moved to the preparation assembly **24** that includes chambers/baths **24***a*-**24***d* as shown in FIG. **5**F. Aliquots of the collected sample may then be deposited into the various chambers/baths **24***a*-**24***d* as outlined above, with FIG. **5**G illustrating an initial aspiration probe **22** extension into the preparation chamber **24***a*.

(47) An advantage of the sampling system **10** is that when multiple sample analyzers are desirably connected or when more reaction or preparation chambers **24** are desirably added, current analyzers run out of space due to their configurations. The sampling system allows a possibility to add additional reaction chambers without requiring a wider system, which can take up too much countertop space. By putting the preparation chambers above the samples, vertical space is used, which reduces the overall breadth (e.g., along a z-direction) of the system.

(48) Note that though FIGS. **1**A-**5**G generally show a cam system used to control the position of traverse **14**, various alternative embodiments can utilize different approaches. For instance, traverse **14** can be in connection with an angled channel (e.g., where the angled channel angles upward from left to right). One or more rods are connected to one or more locations along traverse **14**, the one or more rods are positioned within the angled channel, and the one or more rods are connected to a motor to move along the angled channel to control the position of traverse **14**.

(49) Another exemplary sampling system **48** is shown with reference to FIGS. **6**A-**6**D. Unlike traverse **14** of sampling system **10**, sampling system **48** includes an angled traverse **50**. Using angled traverse **50** in place of the mechanical system **16** may reduce the number of working parts and any accompanying maintenance. The angled traverse **50** supports and guides movement of sample aspiration module body **12**, similar to the movement described above. However, in this embodiment, the body **12** moves in an upward direction, along the angled traverse **50**. Similar to sampling system **10**, sampling system **210** includes a sample transport module **30** with cassettes **20***a*, **20***b*. However, the preparation assembly **24** of sampling system **210** is positioned at an upward angle, similar to or otherwise tracking the angled traverse **50**. Accordingly, as the body **12** moves up the traverse **50**, the aspiration probe **22** also moves up the traverse **50**. For the aspiration probe **22** to reach the preparation chambers/baths **24***a*-**24***d*, the preparation chambers/baths **24***a*-**24***d* are similarly positioned (i.e., along a similar z-axial position) at upwardly angled positions which may be the same or different than the angle of angled traverse **50**. For example, the preparation chambers/bath **24***a*-**24***d* may resemble a stairstep or an upwardly staggered configuration.

(50) As shown by FIG. **6**A, the sample aspiration module body **12** is positioned over a first specimen tube **18***a* of specimen tube arrangement **18**. FIG. **6**B illustrates the aspiration probe **22** collecting a sample. The aspiration probe **22** may include a longer needle to access the tube arrangement **18**. Once the sample has been collected and the aspiration probe **22** has been retracted, the sample aspiration module body **12** moves up the angled traverse **50**. Once sample aspiration module body **12** reaches the position illustrated by FIG. **6**C, the aspiration probe **22** extends into the first preparation chamber **24***a*, as shown by FIG. **6**D. The process may proceed as described above and continue in a similar manner for additional samples. After the final sample has been deposited in bath **24***d*, the sample aspiration module body **12** moves back down the angled traverse **50** to collect a sample from the second specimen tube **18***b*. This process may continue in a similar manner for the third, fourth, and fifth specimen tube **18***c*-**18***e*, and so forth with specimen tubes **18***f*-**18***j*. FIGS. **6**A-**6**D show a manual sampling option (using a manual sampling holder **40**) and an

open vial manual option (using a holder **44**), which may be accessed similarly to the process outlined above.
(51) Alternatively, aspiration probe **22** is not retractable and instead may be elongated to reach the preparation assembly **24**. In one example, sample aspiration module body **12** moves independently along a y-direction (i.e., vertically) relative to angled traverse **50**. In another example, aspiration module body **12** is fixed along a y-direction (i.e., vertically) relative to angled traverse **50** but angled traverse **50** can tilt to control the position of aspiration probe **22**.
(52) It should be understood that various different features described herein may be used interchangeably with various embodiments. For example, if one feature is described with respect to particular example, it is understood that that same feature may be used with other examples as well.
(53) Although certain embodiments have been shown and described, it should be understood that changes and modifications, additions and deletions may be made to the structures and methods recited above and shown in the drawings without departing from the scope or spirit of the disclosure or the following claims.

Claims

1. A blood analyzer sampling system comprising: a plurality of specimen tubes; a traverse; an aspirator configured to aspirate a sample from a specimen tube from the plurality of specimen tubes, travel along the traverse, and deposit the sample into at least one preparation chamber for processing of the sample; wherein the plurality of specimen tubes each have bases which are vertically offset from a base of the at least one preparation chamber and are along a same plane, which plane includes the aspirator, the plurality of specimen tubes and the at least one preparation chamber; and wherein vertically offset means offset in a direction of gravitational acceleration toward earth's center.
2. The sampling system of claim 1, further comprising a cassette containing the plurality of specimen tubes and a mixer configured to receive the cassette and mix the plurality of specimen tubes.
3. The sampling system of claim 1, wherein the at least one preparation chamber is positioned above the plurality of specimen tubes.
4. The sampling system of claim 1, wherein the traverse, the aspirator, the plurality of specimen tubes, and the at least one preparation chamber all lie in a sample plane.
5. The sampling system of claim 1, further comprising a sample transporter configured to move the plurality of specimen tubes, and a mixer, wherein the mixer is configured to mix the plurality of specimen tubes out of a line of travel of the sample transporter.
6. The sampling system of claim 1, wherein aspirator comprises an aspiration probe which is both extendable and retractable.
7. The sampling system of claim 1, wherein; the aspirator is operable in connection with the traverse to gather the sample from a specimen tube of the plurality of specimen tubes at a first position, and to deposit the sample into a preparation chamber from the at least one preparation chamber at a second position; and both the first position and the second position lie in the same plane.
8. The sampling system of claim 1, wherein: the aspirator is operable in connection with the traverse to gather the sample from a specimen tube of the plurality of specimen tubes at a first position, and to deposit the sample into a preparation chamber from the at least one preparation chamber at a second position; and the first position has a different horizontal and vertical location than the second position.
9. The sampling system of claim 1, wherein the at least one sample preparation chamber comprises two or more sample preparation chambers, each in a vertically staggered configuration.
10. A blood analyzer sampling system comprising: a plurality of specimen tubes; a traverse; an

aspirator in operable connection with the traverse; the traverse configured to move from a first position where the aspirator is configured to gather a sample from a specimen tube from the plurality of specimen tubes to a second position where the aspirator is configured to deposit the sample into a preparation tube for processing of the sample; wherein the plurality of specimen tubes each have bases which are vertically offset from a base of the preparation chamber and are along a sample plane, which plane includes the aspirator, the plurality of specimen tubes, and the preparation chamber; and wherein vertically offset means offset in a direction of gravitational acceleration toward earth's center.
11. The sampling system of claim 10, wherein the traverse is connected to two cams which are configured to move the traverse from the first position to the second position.
12. The sampling system of claim 10, further comprising a cassette containing the plurality of specimen tubes and a mixer configured to receive the cassette and mix the plurality of specimen tubes.
13. The sampling system of claim 10, wherein the first position has a different horizontal and vertical location than the second position.
14. The sampling system of claim 10, further comprising a sample transporter configured to move the plurality of specimen tubes, and a mixer, wherein the mixer is configured to mix the plurality of specimen tubes out of a line of travel of the sample transporter.
15. A hematology sampling system comprising: a plurality of specimen tubes and a plurality of preparation chambers; an aspirator configured collect a sample from one of the plurality of specimen tubes at a first position and deposit the sample in at least one of the plurality of preparation chambers at a second position; wherein the plurality of specimen tubes each have bases which are vertically offset from a base of the at least one preparation chamber and are along a same plane which includes the plurality of specimen tubes, the aspirator and the at least one preparation chamber; and wherein vertically offset means offset in a direction of gravitational acceleration toward earth's center.
16. The sampling system of claim 15, further comprising a traverse in operable connection with the aspirator.
17. The sampling system of claim 16, wherein the aspirator is configured to travel along a traverse between the first position and the second position.
18. The sampling system of claim 16, wherein the traverse is configured to move to position the aspirator at the first position and at the second position.
19. The sampling system of claim 15, wherein the aspirator comprises an aspiration probe which is both extendable and retractable.
20. The sampling system of claim 15, wherein the plurality of preparation chambers are in a vertically staggered configuration.